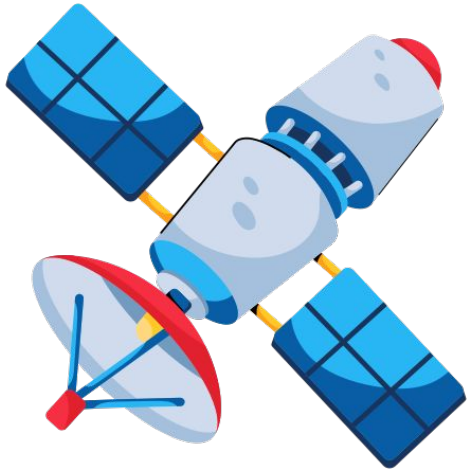


RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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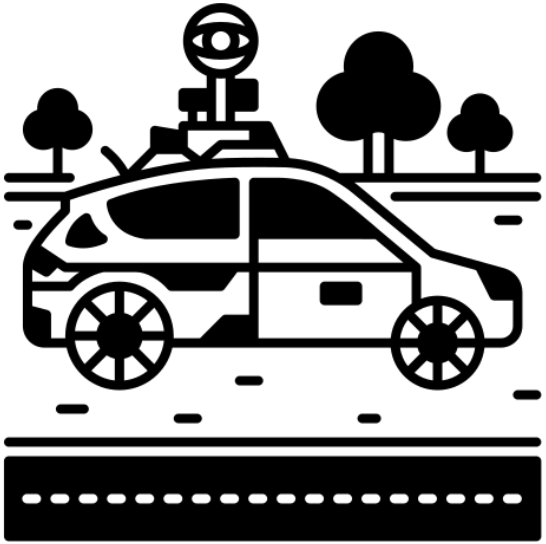
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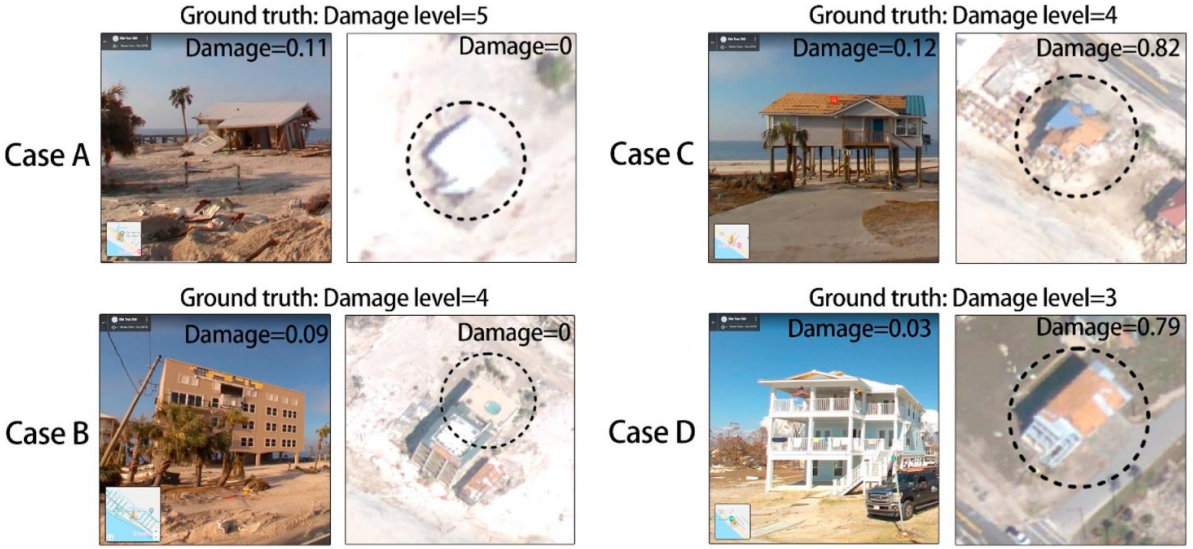




Satellite Imagery--Rapid, large-scale observation immediately after disasters

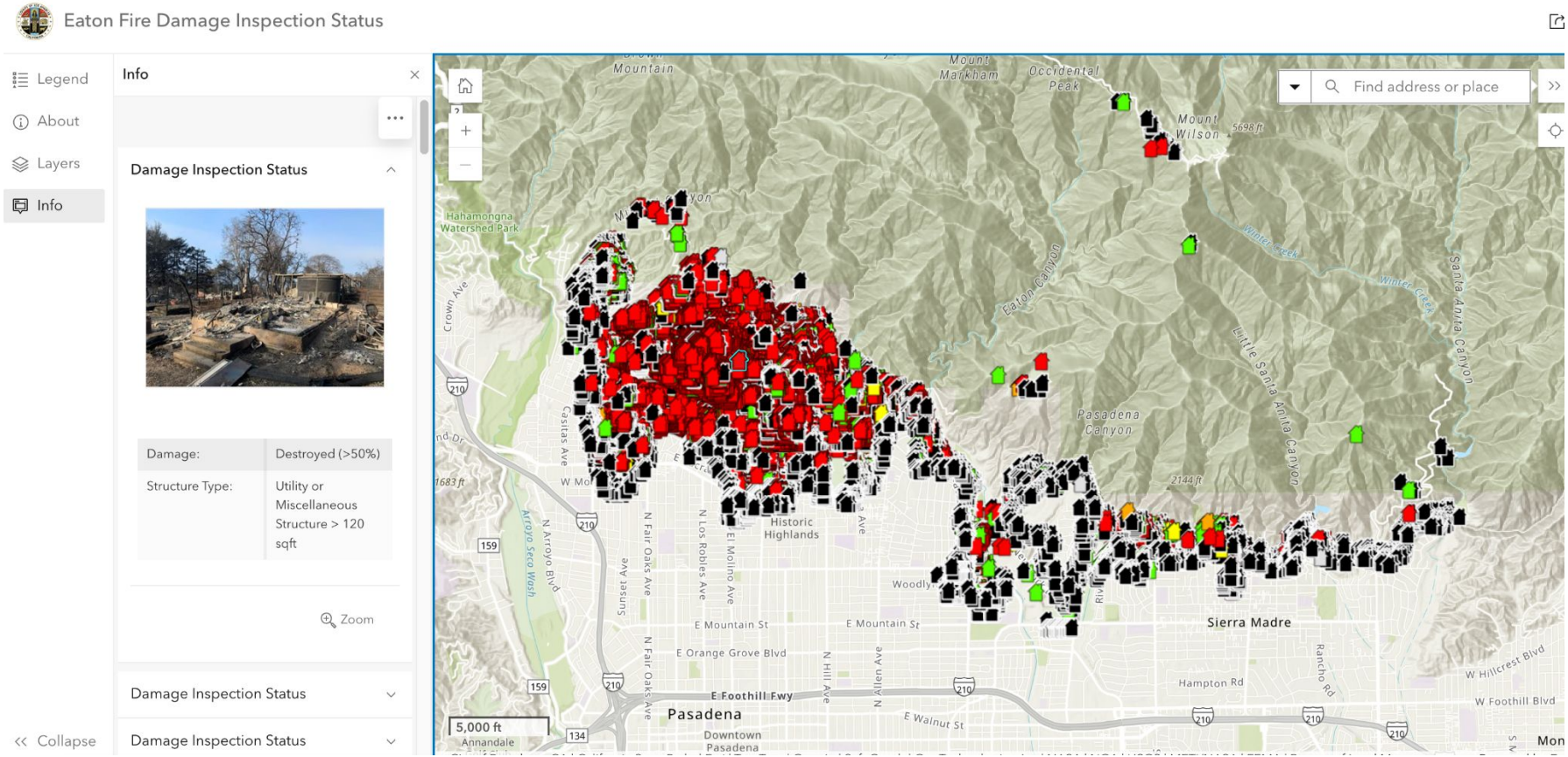
Background





Street-View Imagery-- Human-scale context, essential for structural damage assessment

Background

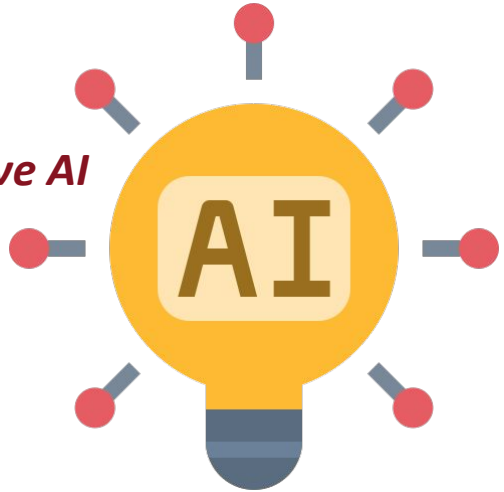


https://lacounty.maps.arcgis.com/apps/instant/sidebar/index.html?appid=2209ecd140d8456291aeba7adbdfd69b

Background

Two Families of AI — (Discriminative vs. Generative)

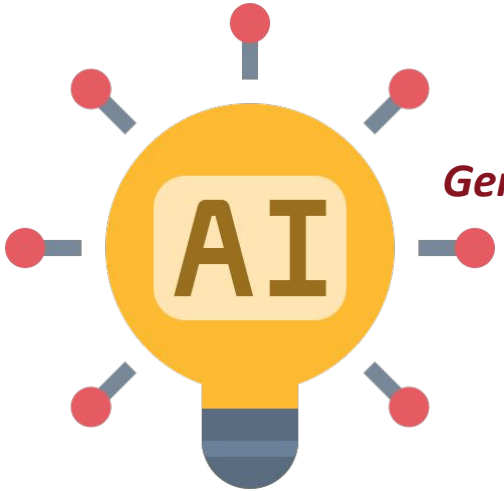
Discriminative AI



Designed for **closed-ended questions with labeled answers** (such as "what is it" or "to what extent"):

"Is this building damaged?"
"What is the extent of the damage—none/minor/severe/total destruction?"

Generative AI



Open-ended/Constructive Questions ("What is missing? / What if...? / How should it be presented, and why?")

"Why was this location assessed as severely damaged? What actions should rescue personnel take here?"

Research Question



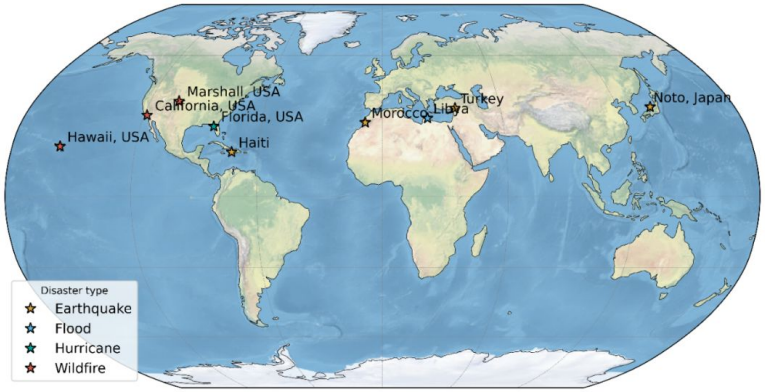
Research Question

Can RAPIDmap support:

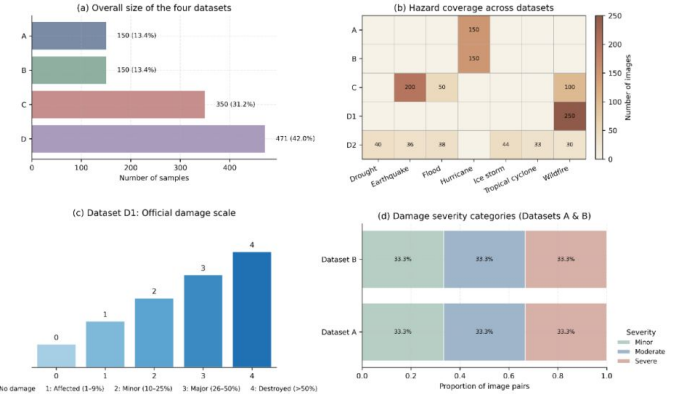
- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?

Dataset

a. Geographic location of events



b. Key statistics



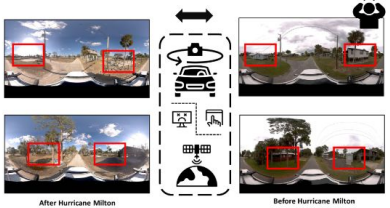
c. Dataset description

Dataset A: Post-disaster street-view and remote-sensing image pairs



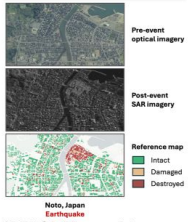
It sourced from the CVDisaster dataset.
It provides paired street-view and remote-sensing imagery from hurricane-affected regions, each annotated with three levels of structural damage (light, medium, heavy).

Dataset B: Pre- and post-disaster street-view image pairs



It consists of bitemporal street-view image pairs from the BiTemporal StreetView Damage dataset. Each pair includes precise location metadata and corresponding severity labels (minor, moderate, severe), mainly collected from Hurricane-impacted area.

Dataset C: Pre- and post-disaster remote-sensing image pairs

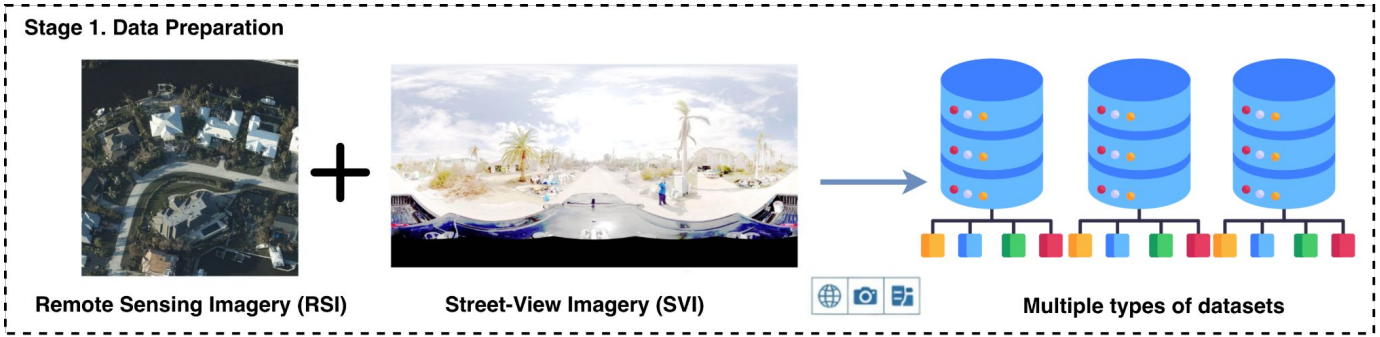


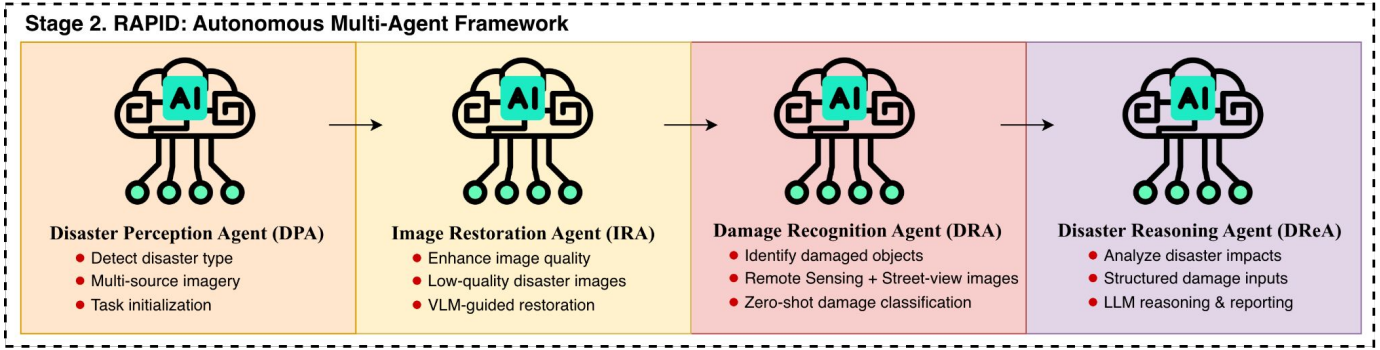
It (BRIGHT) provides pre-/post-disaster satellite imagery with pixel-level building-damage masks, covering multiple hazard types

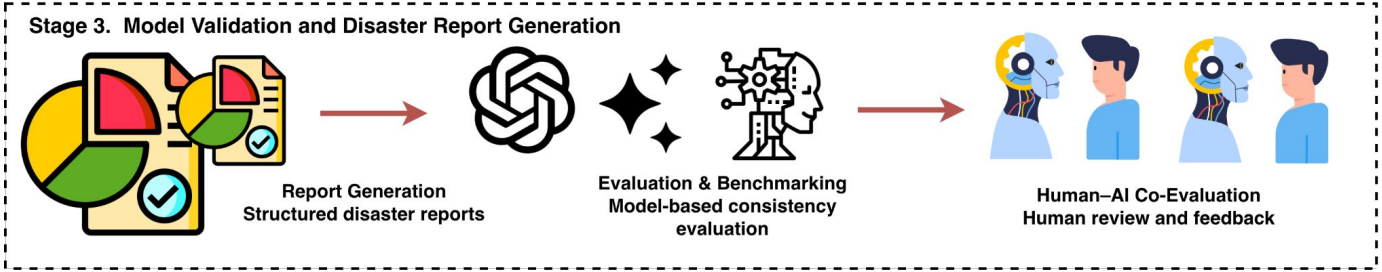
Dataset D: Post-disaster street-view images



It consists of post-disaster ground-level images, including geo-referenced wildfire inspections (D1) and multi-hazard web-sourced scenes without geolocation (D2).







Method

- Cross-view imagery input
- Zero-shot multi-agent assessment
- Structured report generation

Components of the Multi-Agent Framework: Disaster Perception Agent



3.1.1 Disaster Perception Agent. DPA serves as the entry point of the pipeline. We use perception in the computer-vision sense of visual identification and scene parsing, not in the disaster-studies sense of human risk perception. DPA contains ModePerceiver, DisasterReasoner, and TaskPlanner. Together, they infer the input modality, disaster category, restoration need, and downstream tool route from a single multimodal prompt.

We define the perception state as

$$\mathbf{z}_p = (\hat{m}, \hat{d}, \hat{\mathbf{r}}, \hat{c}, \tau), \qquad \tau = \{\text{DRA}, \text{DReA}\} \cup \{\text{IRA} : \|\hat{\mathbf{r}}\|_1 > 0\}, \quad (2)$$

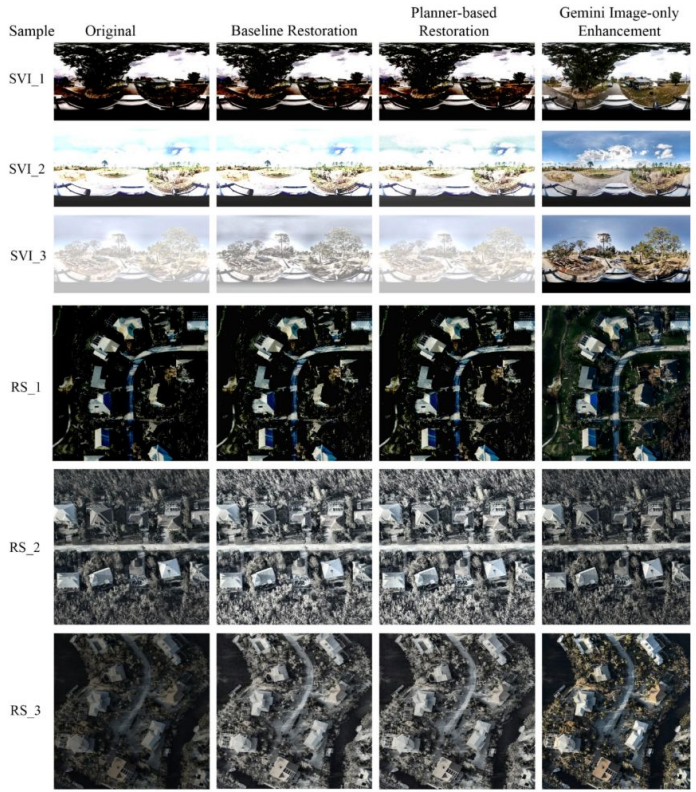
where $\hat{m} \in \mathcal{M}$ is the modality, $\hat{d} \in \mathcal{D}$ is the disaster category, $\hat{\mathbf{r}} \in \{0, 1\}^n$ marks images requiring restoration, $\hat{c}$ is confidence, and τ is the planned agent route. This compact state replaces repeated free-form handoffs with a consistent routing interface.

Components of the Multi-Agent Framework: Disaster Perception Agent

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Model		Drought	Earthquake	Flood	Hurricane	Ice storm	Wildfire
Precision ↑	GPT-5.1-mini	0.95	0.91	0.90	0.98	1.00	0.99
	GPT-5.1	0.97	0.95	0.92	0.97	0.97	0.99
	Gemini-2.5-flash	0.95	0.95	0.90	0.91	1.00	0.99
Recall ↑	GPT-5.1-mini	0.93	0.58	0.97	0.99	0.86	0.98
	GPT-5.1	0.85	0.56	0.92	1.00	0.70	0.97
	Gemini-2.5-flash	0.93	0.50	0.92	1.00	0.50	0.99
F1-score ↑	GPT-5.1-mini	0.94	0.71	0.94	0.99	0.93	0.98
	GPT-5.1	0.91	0.70	0.92	0.99	0.82	0.98
	Gemini-2.5-flash	0.94	0.65	0.91	0.95	0.67	0.99
Overall Accuracy ↑	GPT-5.1-mini				0.92		
	GPT-5.1				0.88		
	Gemini-2.5-flash				0.86		

Components of the Multi-Agent Framework: Image Restoration Agent



3.1.2 Image Restoration Agent. IRA treats restoration as guarded quality control. For each image marked by DPA, it computes no-reference IQA features for contrast, sharpness, and naturalness. The composite score is

$$Q = \frac{w_c C_{\text{norm}} + w_s S_{\text{norm}} + w_n (1 - N_{\text{norm}})}{w_c + w_s + w_n}. \quad (3)$$

Here, C_{norm}, S_{norm}, and N_{norm} denote normalized contrast, sharpness, and naturalness-deviation terms. Higher Q indicates better perceptual quality for downstream interpretation. In the reported experiments we use $w_c = 0.4$, $w_s = 0.4$, and $w_n = 0.2$, prioritizing contrast and sharpness while still penalizing unnatural image statistics. These weights are fixed across datasets rather than tuned per disaster type.

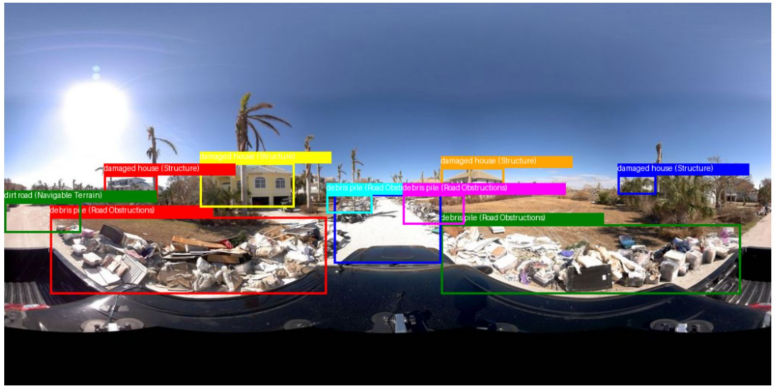
Components of the Multi-Agent Framework: Image Restoration Agent

Category	Total amount	Disaster type	Image type	Number of restoration	$Q_{original}$	$Q_{baseline}$	$Q_{planner}$	Q_{gemini}
A (Post-disaster SVI + RSI pairs)	150	Hurricane	Satellite	141	0.62	0.73	0.71	0.69
		Hurricane	SVI	22	0.75	0.78	0.76	0.79
B (Pre/Post SVI pairs)	150	Hurricane	SVI	4	0.76	0.78	0.78	0.79
C (Post-disaster SVI)	471	Wildfire	SVI	14	0.61	0.67	0.67	0.57
		Tropical	SVI	2	0.51	0.55	0.62	0.60
		Ice storm	SVI	4	0.73	0.74	0.75	0.67
		Flooded	SVI	2	0.68	0.73	0.77	0.71
		Earthquake	SVI	2	0.71	0.79	0.80	0.77
		Drought	SVI	2	0.66	0.75	0.78	0.61
	295	Wildfire	SVI	3	0.70	0.77	0.78	0.78





Components of the Multi-Agent Framework: Damage Recognition Agent





3.1.3 Damage Recognition Agent. DRA converts the audit-preserving image set into ordinal severity and object-level damage indicators. It supports cross-view RSI+SVI inputs, bi-temporal SVI pairs, and single post-disaster SVI. For K severity levels and J damage indicators, let $\pi_{1:K} \in [0,1]^K$ denote severity confidences and $\alpha_{1:J} \in [0,1]^J$ object confidences. The decision rule is

$$\hat{y} = \arg\max_{k \in \{0,\ldots,K-1\}} \pi_k, \qquad \hat{o}_j = \mathbb{1}[\alpha_j \geq \theta_j], \tag{5}$$

where θ_j is the threshold for the j-th damage element. The structured DRA state is $\mathbf{z}_d = (\hat{y}, \hat{\mathbf{o}}, \pi_{1:K}, \alpha_{1:J}, \hat{\mathbf{b}}, \hat{\mathbf{p}})$, where $\hat{\mathbf{b}}$ and $\hat{\mathbf{p}}$ are optional bounding boxes and polygon masks. This schema keeps cross-view, bi-temporal, and single-view cases comparable while preserving confidence information for reasoning.

We evaluate DRA with exact severity accuracy, object-level precision, recall, F1-score, and the Normalized Cross-Severity Error (NCSE), where the "S" denotes severity. NCSE complements categorical accuracy by penalizing large ordinal mistakes more strongly than adjacent-class errors.

Components of the Multi-Agent Framework: Damage Recognition Agent

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Table 4: Performance comparison of the Damage Recognition Agent in multi-disaster severity levels.

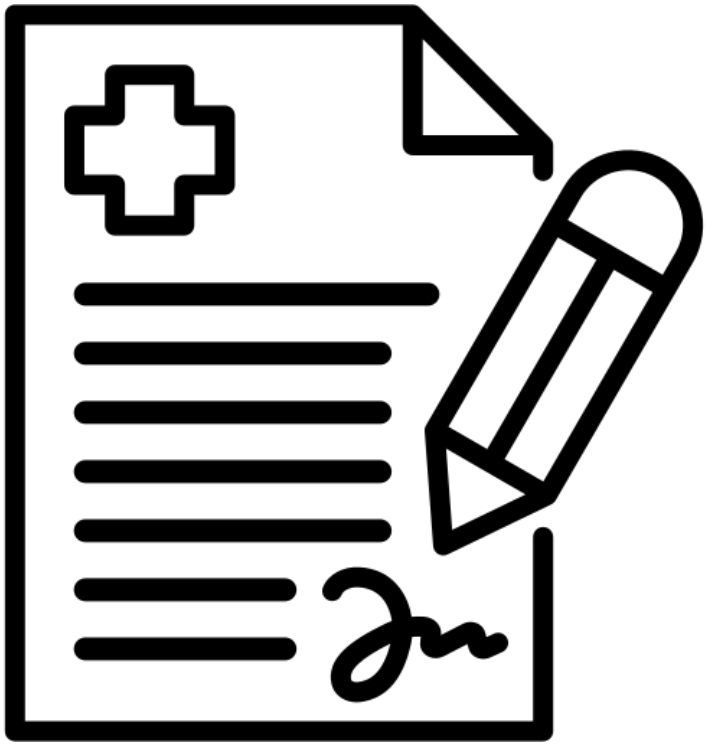
Model		Hurricane (Dataset A)	Hurricane (Dataset B)	Wildfire (Dataset C)
Overall Accuracy ↑	GPT-5-mini	0.387	0.503	0.573
	GPT-5.1	0.573	**0.591**	0.570
	Gemini-2.5-flash	0.360	0.470	0.559
	Gemini-2.5-Pro	0.380	0.447	**0.590**
	Gemini-3-Pro	**0.627**	0.493	0.442
Normalized Cross-Severity Error ↓	GPT-5-mini	0.307	0.248	0.165
	GPT-5.1	0.213	**0.218**	0.173
	Gemini-2.5-flash	0.363	0.299	0.179
	Gemini-2.5-Pro	0.373	0.327	**0.162**
	Gemini-3-Pro	**0.190**	0.291	0.210

Components of the Multi-Agent Framework: Damage Recognition Agent

Table 5: Quantitative performance of different models on disaster damage recognition across cross-view hurricane imagery (Dataset A) and bi-temporal street-view imagery (Dataset B).

Category	Model	F1-score	Recall	Precision	Accuracy	Damaged Building	Debris	Downed Power Line	Fallen Tree	Flooded Area
Hurricane Dataset A (Post-disaster street-view and remote-sensing image pairs)	Gemini-3-Pro	**0.77**	0.78	**0.80**	**0.93**	**0.94**	0.89	**1.00**	**0.92**	0.88
	Gemini-2.5-Pro	0.57	**0.91**	0.50	0.72	0.57	0.77	0.67	0.63	**0.95**
	Gemini-2.5-flash	0.55	0.81	0.54	0.75	0.53	0.76	0.83	0.67	0.94
	GPT-5.1	0.54	0.58	0.64	0.84	0.74	**0.84**	0.97	0.74	0.91
	GPT-5-mini	0.50	0.58	0.46	0.79	0.75	0.81	0.92	0.60	0.90
Hurricane Dataset B (Pre- and post-disaster street-view image pairs)	Gemini-3-Pro	0.39	0.36	0.51	0.85	0.74	0.80	0.99	0.72	0.92
	Gemini-2.5-Pro	0.43	0.47	0.40	0.79	0.72	0.79	0.77	0.69	0.79
	Gemini-2.5-flash	0.43	0.44	0.43	0.83	0.78	0.78	0.93	0.69	0.88
	GPT-5.1	**0.55**	**0.54**	**0.57**	**0.96**	**0.97**	**0.92**	**1.00**	**0.91**	**0.99**
	GPT-5-mini	0.47	0.49	0.45	0.86	0.78	0.82	0.97	0.74	0.79

Components of the Multi-Agent Framework: Disaster Reasoning Agent



3.1.4 Disaster Reasoning Agent. DReA maps structured recognition outputs to explanation and action. It conditions the report on perception state, damage state, image evidence, and a recovery-knowledge prompt κ:

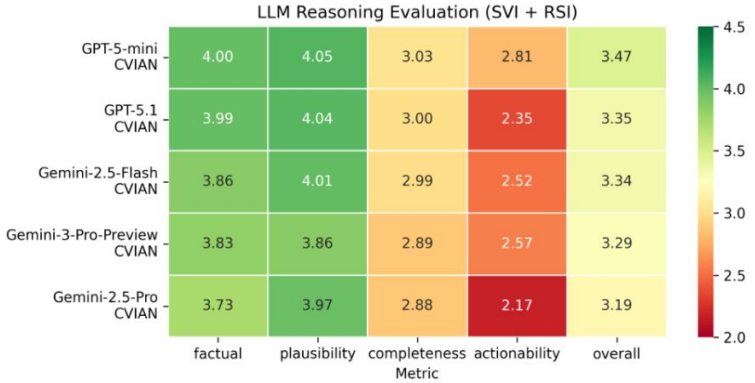
$$\mathcal{Y}_i = A_{\mathrm{DReA}}(\mathbf{z}_{p,i}, \mathbf{z}_{d,i}, \tilde{X}_i, \kappa). \tag{6}$$

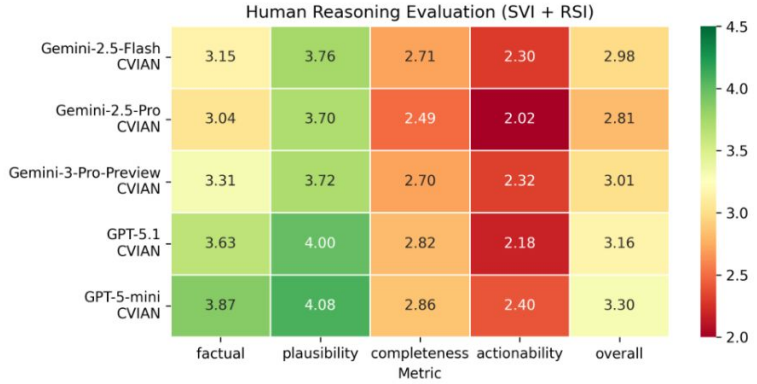
The report describes damage mechanisms, affected assets, secondary risks, and recovery suggestions such as debris clearance, vegetation restoration, structural safety inspection, and community risk mitigation.

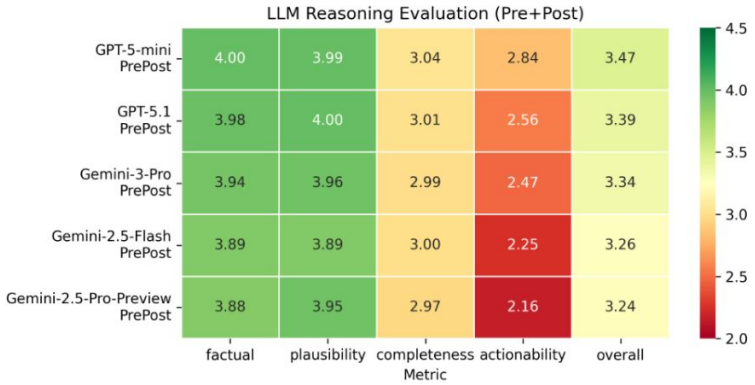
Because $\mathcal{Y}_i$ is open-ended text, we evaluate it with automated and human scores. Let $\mathcal{R}$ denote four rubric dimensions: factual consistency, causal plausibility, completeness, and action feasibility. The automated judge is used as a first-pass rubric scorer rather than as ground truth, because LLM-based evaluation can favor familiar styles of reasoning. For human evaluation, three geography Ph.D. student reviewers independently score the sampled reports using the same 1–5 rubric after inspecting the associated image evidence and generated report. The final human score is the mean of their rubric scores across reviewers and dimensions:

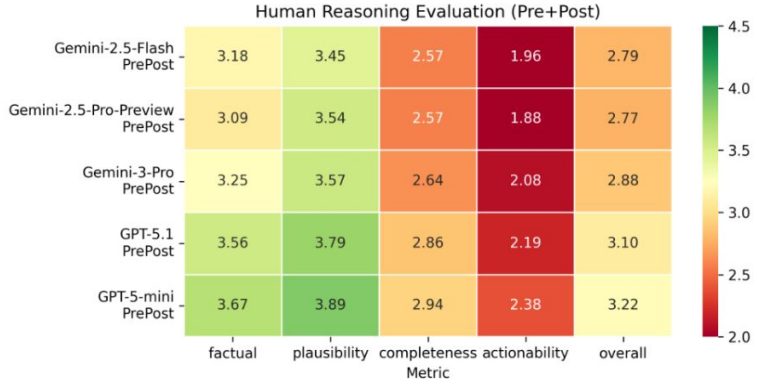
$$S_i^{\mathrm{auto}} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} s_{i,r}^{\mathrm{auto}},$$
$$S_i^{\mathrm{human}} = \frac{1}{3|\mathcal{R}|} \sum_{h=1}^{3} \sum_{r \in \mathcal{R}} s_{i,h,r}^{\mathrm{human}}. \tag{7}$$

Components of the Multi-Agent Framework: Disaster Reasoning Agent



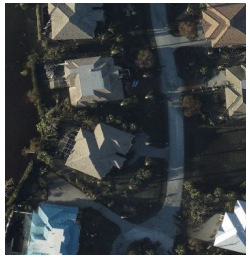


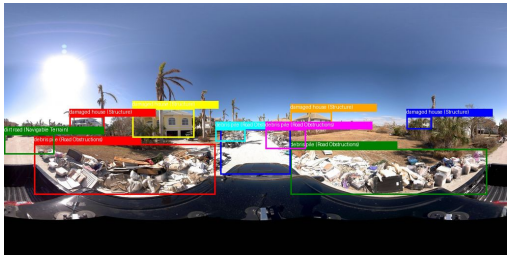




Final output from the disaster-reasoning agent

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Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
 	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building



Using Gemini-3-Pro for object recognition

The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**

Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**

Takeaways

- A zero-shot multi-agent pipeline capable of transforming satellite and street-view imagery into traceable attribute information ready for mapping, while also providing corresponding textual descriptions for each location.

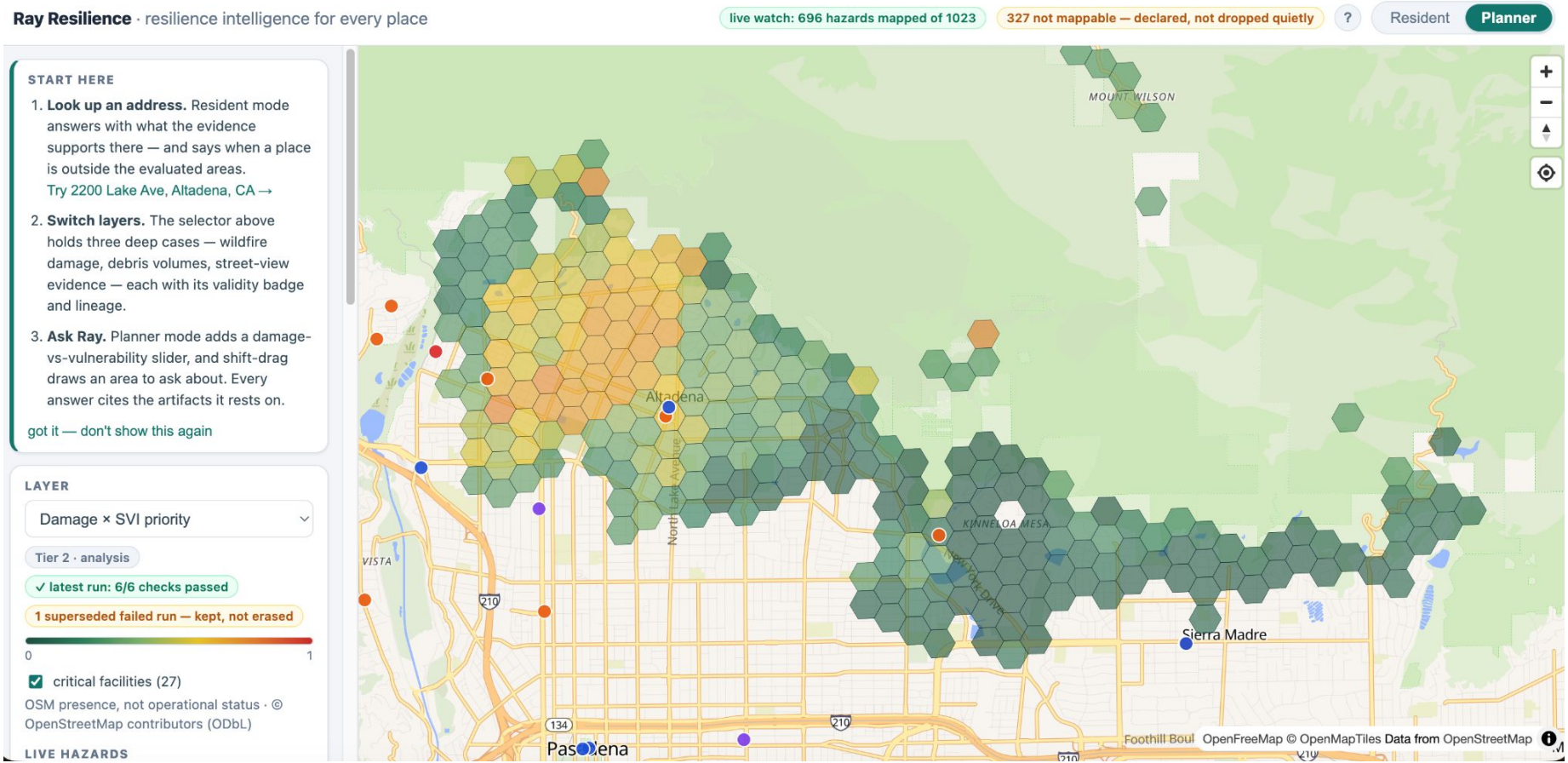
- Severity and confidence are not yet map-grade: severity accuracy stays at 0.38–0.63, self-reported confidences are uncalibrated, and LLM judges rate reports higher than human reviewers do.

- Disaster mapping with agents should be interactive, not one-shot. The agent proposes, explains, and revises; the cartographer questions, corrects, and decides what reaches the map.



Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. In *The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*, November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3841645.3843346

Future Directions





Q&A

Welcome any comments/questions

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